Section 13.1: Vector Functions and Space Curves

Vector Functions

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Vector Function

A **vector function** (or **vector-valued function**) is a function whose input is a scalar t and whose output is a vector:

$$\vec{r}(t) = \langle f(t),\, g(t),\, h(t)\rangle = f(t)\,\mathbf{i} + g(t)\,\mathbf{j} + h(t)\,\mathbf{k}$$

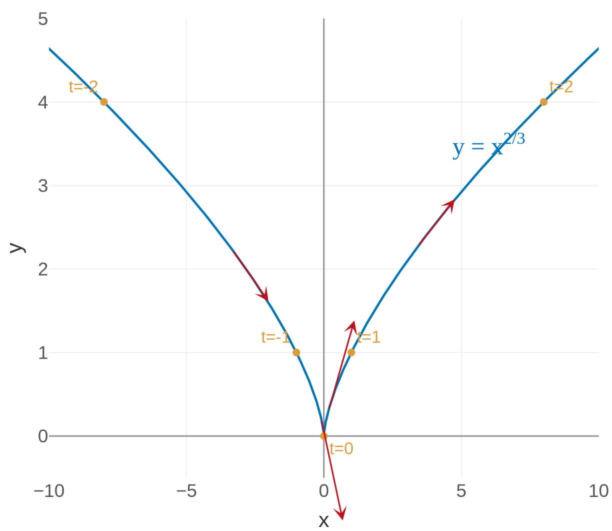
The functions f, g, h are called the **component functions**.

A vector function is just a way of packaging **parametric equations** into a single object.

Why Vector Functions?

Chapter 13 is about **calculus on curves** — the same derivative and integral ideas from Calc I, but now applied to vector functions.

Example 1: Sketch $\vec{r}(t) = \langle t^3,\, t^2\rangle$



Interactive version

The curve $y = x^{2/3}$ has a **cusp** at the origin. The parametrization tells us *which way* to travel along the curve.

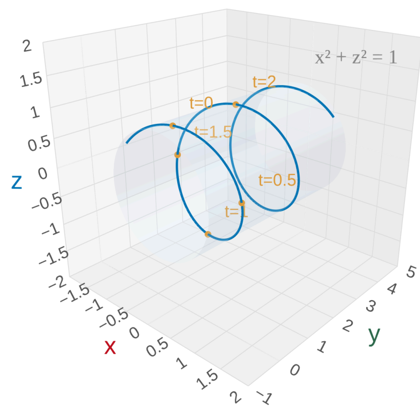
Your Turn: Sketch $\vec{r}(t) = \langle 2\cos t,\, 2\sin t\rangle$

Try it before looking at the solution.

Space Curves in 3D

The component functions are the clues — they tell you what surface the curve lives on and how it moves.

Example 2: Sketch $\vec{r}(t) = \langle \sin \pi t,\ t,\ \cos \pi t \rangle$



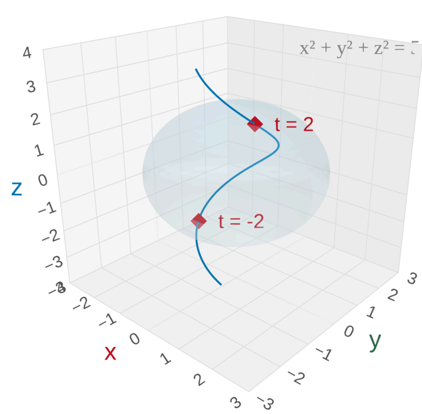
Interactive version

This is a **helix** — trig + trig + linear = spiral. Think of wrapping a string around a paper towel tube.

Where Does a Curve Meet a Surface?

This reduces a 3D geometry problem to **solving an equation in one variable** t.

Example 3: Where does $\vec{r}(t) = \langle \sin t,\ \cos t,\ t \rangle$ meet $x^2 + y^2 + z^2 = 5$?



Interactive version

The helix pierces the sphere at exactly **two points**, at $t = \pm 2$. The Pythagorean identity did the heavy lifting!

Your Turn: Where does $\vec{r}(t) = \langle t,\ 2t,\ 3t \rangle$ meet $x^2 + y^2 + z^2 = 14$?

Try it before looking at the solution.

Homework

Section 13.1: 3, 5, 7, 11, 15, 19, 25–30